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To cite this article: Qing Liu, Yuxia Fu & Chen Zhu (2025) Economic policy uncertainty and bank stability: the role of digitalization level, *Journal of Chinese Economic and Business Studies*, 23:2, 165-190, DOI: [10.1080/14765284.2024.2435226](https://doi.org/10.1080/14765284.2024.2435226)

To link to this article: <https://doi.org/10.1080/14765284.2024.2435226>



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Economic policy uncertainty and bank stability: the role of digitalization level

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ABSTRACT

This paper employs quarterly data from 102 Chinese commercial banks spanning from 2017Q1 to 2021Q3. Utilizing a panel fixed-effects model, we scrutinize the impact of digitization on the relationship between economic policy uncertainty and the stability of commercial banks. The main research conclusion is that a higher level of digitization can alleviate the negative effects of economic policy uncertainty on the stability of commercial banks. Further research indicates that digitization achieves this effect by improving borrower selection, reducing the cost of risk management, and enhancing profitability. The mitigating effect of digitization varies across different types of commercial banks, different time periods, and different external environments, with a stronger impact observed on state-owned banks, banks in the post-pandemic period, and banks operating in high external pressure environments. Our research results demonstrate for the first time how digitization enhances banks' resilience to the impact of economic policy uncertainty, providing insights for financial stability and development.

ARTICLE HISTORY

Received 5 August 2023

Accepted 16 November 2024

KEYWORDS

Economic policy uncertainty; bank stability; digitalization level

1. Introduction

In today's era of globalization and digitalization, development issues and financial development have become one of the focal points of international attention (Beck et al. 2023; Suri, Bharadwaj, and Jack 2021). However, developing countries face numerous challenges in achieving sustainable development, including the impacts of economic policy uncertainty (Phan et al. 2021; Rodrik 1991). Against this backdrop of uncertainty, commercial banks, as core institutions of the financial system, experience significant impacts on their stability (Nguyen 2021; Wu et al. 2020, 2022) and operational development (Berger et al. 2022; Quagliariello 2009), thereby affecting the stability of financial markets and the prospects for economic development.

As the most significant technological innovation since the 21st century, digital technology has transformed the banking industry, and a series of uncertain events (such as the outbreak of the COVID-19 pandemic) have accelerated this trend, making digitization a strategic priority for commercial banks (Cappa et al. 2021; Feyen et al. 2021). In this context, delving deeper into the relationship between economic policy uncertainty and the stability of commercial banks, considering the influence of the digitalization level of

commercial banks, holds important theoretical and practical significance. By investigating this issue, we can better understand the challenges and opportunities faced by commercial banks in the digital age, provide valuable insights for the formulation of effective policies and management measures, promote the robust development of the financial system, and foster long-term economic growth and sustainable development.

The practical significance of this paper is particularly relevant to the 17 Sustainable Development Goals (SDGs), especially Goal 8 (Decent Work and Economic Growth) and Goal 9 (Industry, Innovation, and Infrastructure). On one hand, the stability of the financial system is closely linked to the real economy, playing a crucial role in ensuring employment and achieving economic growth. On the other hand, the digitalization of commercial banks represents a key form of innovation, which is of great importance for sustainable development. This paper not only offers a new perspective on the interaction between financial stability and economic policy uncertainty but also provides empirical evidence on financial system stability, innovation, and sustainable development. While substantial research has explored the relationship between economic policy uncertainty and bank stability, few studies have analyzed this relationship from the perspective of digitalization levels. This paper fills this gap, aiming to provide new evidence for policymakers, bank managers, and academics to support the digital transformation of the banking sector under uncertain policy environments and contribute to the broader achievement of sustainable development goals.

The literature relevant to this paper falls into two main streams. Firstly, existing research primarily focuses on the influence of institutional quality, regulation and supervision, political connections, and executive power on the relationship between uncertainty and commercial banks. Previous research has shown that uncertainty increases the risk and reduces the stability of commercial banks (Wu et al. 2020), but higher institutional quality (Shabir et al. 2021), stringent capital regulation, and official regulatory authority (Nguyen 2021) can mitigate this adverse effect. Additionally, banks tend to adopt a more conservative lending approach and exhibit a decreased willingness to take proactive risks when facing high levels of economic policy uncertainty. However, when banks have political connections, this adverse impact is mitigated, leading to a moderation of the negative effects (H. Cheng et al. 2021). Shabir et al. (2023) also found that increasing economic and geopolitical uncertainty significantly deteriorates bank stability, while CEO power and a robust board of directors enhance bank performance and minimize the adverse impact of economic and geopolitical uncertainty on bank risk. This paper contrasts sharply with the extensive research on factors influencing the relationship between economic policy uncertainty and bank stability (or risk). Departing from existing research, this paper focuses on the own characteristics of commercial banks, examining whether their level of digitization can mitigate the adverse impact of economic policy uncertainty on bank stability – a question that remains insufficiently explored.

Moreover, the current understanding of the impact of digitalization level on bank stability under economic policy uncertainty is unclear, with much of the literature focusing on banking informatization and fintech. On one hand, studies on banking informatization have found that higher levels of information technology adoption can reduce banks' non-performing loan ratios during financial crises (Pierri and Timmer 2022), lead to better performance of highly informatized banks during pandemics (Dadoukis, Fiaschetti, and Fusi 2021), enable more lending (Branzoli, Rainone, and Supino 2024), and enhance

banks' capacity to serve customers and attract more deposits (Kwan et al. 2021). Regarding fintech, literature suggests that banks' fintech capabilities can decrease credit risks (M. Cheng and Qu 2020), improve bank performance, and have a positive impact on the overall industry (W. Chen and Srinivasan 2023). On the other hand, digitalization may bring about various issues and challenges. Stoica, Mehdian, and Sargu (2015) found that information technology does not necessarily improve banks' operational efficiency. The diversification of products, services, and business brought about by digital technology may further exacerbate the negative impact of economic policy uncertainty, as non-interest income exhibits high instability during periods of increased uncertainty (Demirgüç-Kunt and Huizinga 2010; Stiroh and Rumble 2006). Therefore, the influence of digitalization level on the relationship between economic policy uncertainty and bank stability remains to be empirically studied.

For developing countries, the level of economic policy uncertainty tends to be higher (Bloom 2014), making it an important question to explore whether the level of digitalization can influence the relationship between economic policy uncertainty and the stability of commercial banks. Existing research on the heterogeneous impact of digitalization on bank performance under uncertainty has primarily focused on developed countries, such as the United States (Pierri and Timmer 2022) and Italy (Branzoli, Rainone, and Supino 2024). In developing countries like China, further investigation is needed into the impact of bank digitalization level. China, as one of the rapidly developing countries in terms of bank digitalization,¹ provides an excellent case for analyzing bank digitalization and its related effects.

This paper utilizes quarterly panel data from Chinese commercial banks to empirically analyze the impact of digitalization level on the relationship between economic policy uncertainty and bank stability. The results reveal the following: First, the level of digitalization effectively mitigates the negative impact of economic policy uncertainty on bank stability. This conclusion remains significant even after controlling for various bank-level and macro-level variables. The reliability of this finding is not affected by variable replacements or addressing endogeneity concerns. Second, mechanism analysis indicates that the mitigating effect of digitalization is mainly achieved through borrower screening, reducing the management costs of risk management, and enhancing profitability. Third, heterogeneity analysis at different levels shows that the mitigating effect of digitalization is more pronounced in non-state-owned banks, during the post-pandemic period, and in provinces where banks face higher external competitive pressures.

The contribution of this paper is primarily reflected in three aspects: First, this paper innovatively measures the digitalization level of commercial banks. Digitalization in banks differs from informatization or fintech. As emphasized by Pierri and Timmer (2022), the information technology of the early 21st century is entirely distinct from today's digitalization, which brings new strategic planning and business processes to banks. Fintech typically refers to the utilization of advanced technology and innovative techniques to change and optimize traditional financial services, covering a broad range of fields such as payment services, lending platforms, investment management, and insurance technology (Gomber et al. 2018). In contrast, the digitalization of commercial banks focuses more on the digital transformation of banking operations (Wadesango and Magaya 2020). Therefore, this paper constructs a digitalization level index of banks to conduct more comprehensive research and analysis. We innovatively utilize web crawling technology

and use the quantity of news articles from Baidu² News to measure the digitalization level of commercial banks. In comparison with the approach of M. Cheng and Qu (2020) that selects fintech keywords, this paper constructs the digitalization level index from five dimensions: strategic planning, organizational structure, business processes, operational services, and data governance. By focusing on the digital business of banks, this paper comprehensively and multi-dimensionally measures the digitalization level of commercial banks, providing a more direct and accurate reflection of the digitalization level of each bank. This serves as a further complement to the measurement of digitalization level at the micro-level of commercial banks.

Second, this paper extends the existing mechanisms of how digitalization level affects the relationship between uncertainty and bank stability. Unlike previous studies focusing on mechanisms such as customer relationships (Dadoukis, Fiaschetti, and Fusi 2021), online lending services (Branzoli, Rainone, and Supino 2024), and customer acquisition and services (Kwan et al. 2021), this paper explores the impact channels of digitalization level based on the characteristics of digital banks, including digital credit assessment, digital information analysis, and digital service operations. Specifically, we explore the impact channels of digitalization from three perspectives: borrower screening, risk management cost reduction, and profitability. To the best of our knowledge, this paper is the first to simultaneously discuss the impact channels of digitalization from these three aspects. Additionally, we conduct heterogeneous analysis at different levels to further validate the mechanisms of digitalization's effects.

Last, building upon existing research on the relationship between economic policy uncertainty and commercial bank stability, this paper introduces the digitalization level of commercial banks, offering a fresh perspective on the benefits of digitalization in commercial banking. Prior studies have elucidated the impact of economic policy uncertainty on commercial bank stability (Nguyen 2021; Wu et al. 2020, 2022), but have not considered the potential influence of digitalization level on this relationship. By incorporating the digitalization level of commercial banks, this paper expands the understanding of this relationship, providing a new perspective on how commercial banks respond to economic policy uncertainty. This contributes to a deeper understanding of the operations and coping strategies of commercial banks in uncertain environments, offering more comprehensive and practical guidance for the robust development of the banking industry.

The subsequent organization of this paper is as follows: In Section 2, the research hypotheses are proposed through theoretical analysis. Section 3 explains the data sources and empirical models used in this paper. Section 4 presents the results of the baseline regression, robustness tests, and heterogeneity analysis. Finally, Section 5 provides the conclusion of this paper.

2. Theoretical analysis and hypotheses

Previous literature has demonstrated that economic policy uncertainty can lead to poorer bank stability performance (Nguyen 2021; Wu et al. 2020, 2022). In theory, the level of digitalization can influence bank stability by engaging in pathways between the two, as illustrated in Figure 1. Enhanced borrower screening capabilities, reduced risk management costs, and improved profitability are crucial, as they directly relate to the stability

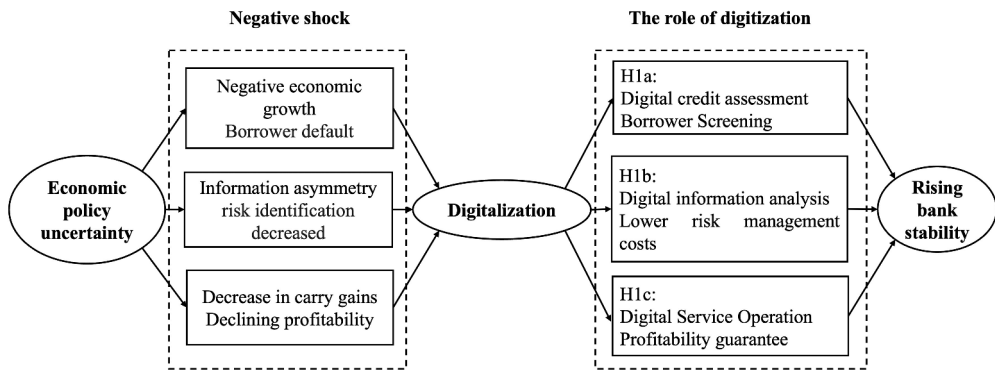


Figure 1. Mechanism of the digitalization level.

and operational performance of commercial banks in response to economic policy uncertainty. Through the influence of digitalization, banks can better cope with changes in the external environment, efficiently manage risks, and maintain stable operations.

First, the level of digitalization can enhance the borrower screening capability of commercial banks through digital credit assessment, thereby alleviating the negative impact of economic policy uncertainty on bank stability. Economic contraction under uncertainty (Fatás and Mihov 2013) increases the probability of borrower default, further translating into increased bank risk and decreased stability. Reduced innovation activities (Bhattacharya et al. 2017) and productivity (Bachmann, Elstner, and Sims 2013) under economic policy uncertainty lead to decreased profitability, resulting in increased default rates for enterprises and decreased stability for commercial banks. However, the level of digitalization can enhance the borrower screening capability of commercial banks (Pierri and Timmer 2022), improve credit quality, and reduce non-performing loan ratios. By leveraging digital technology for credit assessment (Branzoli, Rainone, and Supino 2024), banks can predict the willingness and ability of borrowers to repay loans, reduce the probability of borrower defaults, and consequently enhance the stability of commercial banks.

Second, the level of digitalization can reduce the management costs of risk management for commercial banks through digital information analysis, thereby mitigating the negative impact of economic policy uncertainty on bank stability. Economic policy uncertainty exacerbates information asymmetry (Wu et al. 2020), disrupting bank decision-making and leading to a decline in the bank's risk identification capability. It becomes challenging to accurately predict future returns on investment projects, resulting in weakened risk management and a decrease in stability. Due to heightened uncertainty and information problems, banks' decisions may exhibit herding behavior (Quagliarello 2009), where they imitate the lending decisions of the 'first-mover' banks. Deviating from the bank's 'fundamentals' in such cases may lead to higher risk (Wu et al. 2020). Digitalization can address critical information asymmetry issues (Lapavitsas and Dos Santos 2008), reducing search costs and empowering banks with information collection and analysis capabilities (Cenni et al. 2015). Research by Norden, Buston, and Wagner (2014) from the perspective of bank risk management showed significant benefits of continuous financial innovation under adverse conditions. Banks actively embracing

digital technology demonstrate better risk management performance by relying on computationally intensive processes, enabling quantitative estimation of various financial risks (Lapavitsas and Dos Santos 2008). This helps to reduce the management cost of bank risk management, strengthen risk management capabilities under uncertainty, and reduce the occurrence of homogenous decision-making behavior, thereby safeguarding the stability of commercial banks.

Third, digitalization can enhance the profitability of commercial banks through digital service operations, thereby mitigating the negative impact of economic policy uncertainty on bank stability. Uncertainty leads to a reduction in bank interest spreads and profitability (Wu et al. 2020). However, critically, digital technology enables banks to explore new sources of profits (Lapavitsas and Dos Santos 2008). Digitalization changes banks' business models, reducing indirect expenses (Hernando and Nieto 2007), which in turn enhances the bank's profitability. Beccalli (2007) utilized data from European banks and found that information technology investment positively influences accounting profits and profit efficiency. Therefore, the digitalization level of commercial banks can alleviate the decline in their profitability and mitigate the inhibiting effect of economic policy uncertainty on bank stability.

Thus, this paper proposes the following research hypotheses:

Hypothesis 1 (H1): The level of digitization can mitigate the negative impact of economic policy uncertainty on commercial bank stability.

Hypothesis 1a (H1a): In the face of exacerbated economic policy uncertainty, digitalization level can enhance the borrower selection capability of commercial banks, thereby mitigating the adverse impact of economic policy uncertainty on the stability of commercial banks.

Hypothesis 1a (H1b): In the face of intensified economic policy uncertainty, the level of digitization can reduce the management costs of risk management for commercial banks, thereby alleviating the negative impact of economic policy uncertainty on the stability of commercial banks.

Hypothesis 1a (H1c): In the face of heightened economic policy uncertainty, the level of digitization can enhance the profitability of commercial banks, thereby mitigating the negative impact of economic policy uncertainty on the stability of commercial banks.

3. Research design and data

3.1. Research design

This paper takes the stability of commercial banks as the dependent variable and introduces the interaction term of digitalization level and economic policy uncertainty. The following baseline model is constructed:

$$\ln z_{i,t} = \beta_0 + \beta_1 \text{cnepu}_t + \beta_2 \text{digital}_i + \beta_3 \text{cnepu}_t \times \text{digital}_i + \beta_4 X_{i,t} + \beta_5 M_t + \sigma_i + \varepsilon_{i,t} \quad (1)$$

where the subscripts i and t respectively represent banks and quarters. $cnepu$, $digital$, and lnz represent economic policy uncertainty, commercial bank digitalization level, and stability, respectively. Specifically, this paper limits the search time interval of news articles to the year prior to the regression sample to control for the initial digitalization heterogeneity of commercial banks. Therefore, the variable $digital$ varies only with individual commercial banks. X represents bank-level control variables, and M represents macro-economic control variables. Following H. C. Huang, Wang, and Xiong (2022), due to economic policy uncertainty being time-series data, it is not feasible to control for time fixed effects. Therefore, a series of macro-level control variables are employed. Additionally, for robustness considerations to prevent potential unobserved time-varying omitted variables from affecting the regression results, this paper employs a double fixed-effects model. σ_i denotes bank-specific fixed effects (during regression, $\beta_2 digital_i$ will be absorbed by bank-specific fixed effects), $\varepsilon_{i,t}$ represents the random error term, and β_3 is the coefficient of the core explanatory variable of interest in this paper. To control for potential heteroscedasticity and serial correlation issues, all regressions in this paper use bank-level cluster-robust standard errors.

Following Hainmueller, Mummolo, and Xu (2019), we carefully interpret the coefficients in the empirical regression with interaction terms. The marginal effect of economic policy uncertainty on the stability of commercial banks is:

$$\frac{\partial lnz}{\partial cnepu} = \beta_1 + \beta_3 digital \quad (2)$$

However, we are unable to interpret β_1 as the change in the dependent variable by one unit when the independent variable changes by one unit, given that the moderator variable is set to 0. To facilitate the interpretation of the coefficient for the key variable $cnepu$, we center the independent variable $cnepu$ and the moderator variable $digital$, that is, we demean the variables. Specifically, we define $cnepu_t^* = cnepu_t - \overline{cnepu}$, where $\overline{cnepu}$ represents the mean of $cnepu_t$. Similarly, $digital_i^* = digital_i - \overline{digital}$, where $\overline{digital}$ denotes the mean of $digital_i$. The regression model is specified as follows:

$$lnz_{i,t} = \beta_0^* + \beta_1^* cnepu_t^* + \beta_2^* digital_i^* + \beta_3^* cnepu_t^* \times digital_i^* + \beta_4 X_{i,t} + \beta_5 M_t + \sigma_i + \varepsilon_{i,t} \quad (3)$$

At this point, the marginal effect of economic policy uncertainty on the stability of commercial banks is:

$$\frac{\partial lnz}{\partial cnepu} = \beta_1^* + \beta_3^* (digital - \overline{digital}) \quad (4)$$

The regression coefficient β_1^* can be interpreted as the change in the dependent variable by β_1^* units for each one-unit change in the independent variable when the moderator variable is held at its mean value. In the context of this paper, β_1^* specifically represents the change in the stability of commercial banks when economic policy uncertainty increases by one unit, assuming the level of digitalization of commercial banks is at its average value. Specifically, for a one-unit increase in economic policy uncertainty, the stability of commercial banks will change by $100 \beta_1^* \%$.³ All subsequent regressions follow a similar interpretation.

3.2. Variable selection

3.2.1. Dependent variables

The dependent variable in this paper is the stability of commercial banks (*lnz*). To assess the impact of the level of digitalization on mitigating the negative effects of economic policy uncertainty on bank stability, the *Z – score* is employed as a measure of bank stability, as commonly used in extensive literature (Demirgüç-Kunt and Huizinga 2010; Laeven and Levine 2009). This variable combines various factors, such as operational stability, profitability, and financial leverage, to provide a comprehensive measure of a bank's stability. The *Z – score* is often interpreted as the inverse of the probability of a bank's bankruptcy, where higher values indicate greater bank stability. The specific formula is as follows:

$$Z_{i,t} = \frac{ROA_{i,t} + EA_{i,t}}{\sigma_i(ROA_{i,t})} \quad (5)$$

In which, *ROA* represents the return on assets, *EA* represents the equity ratio, and $\sigma(ROA)$ represents the standard deviation of return on assets.⁴ Considering its characteristics of being right-skewed, heavy-tailed, and possibly having negative values, we followed the approach of Wu et al. (2020) by adding 1 to the *Z – score*, and then taking its natural logarithm to define *lnz*.

To identify the specific channels through which digitalization level affects commercial bank stability, this paper uses the following variables as dependent variables: (1) non-performing loan ratio (*npl*), which reflects the proportion of loans with default risk. (2) Banking risk management cost (*aego**i*), measured by the ratio of management expenses to operating income, where a smaller value indicates stronger risk management capability. (3) Return on assets (*roa*), representing the profitability of commercial banks, where a higher value indicates stronger profitability. (4) Non-interest income ratio (*nii*), which measures the proportion of non-interest income to operating income. Due to the limited availability of quarterly data, there are certain missing values for *npl* and *nii* variables.

3.2.2. Core independent variables

The first core explanatory variable is economic policy uncertainty (*cnepu*). In this paper, the monthly data on overall economic policy uncertainty in China since 1949 is quantified based on two major mainland Chinese newspapers, namely the 'People's Daily' and the 'Guangming Daily', as done by Davis, Liu, and Sheng (2019). To include more monthly information, the quarterly economic policy uncertainty is measured as the arithmetic average of the three-month index values (divided by 100). To ensure robustness in the analysis, the paper also introduces the monthly economic policy uncertainty index (*epu*) innovatively measured by Baker, Bloom, and Davis (2016) based on the 'South China Morning Post.' Similarly, the quarterly index is calculated as the arithmetic average of the three-month values (divided by 100). Figure 2 illustrates the trends in economic policy uncertainty from the first quarter of 2013 to the first quarter of 2023. The data reveal an overall increasing trend in economic policy uncertainty in recent years, and both types of economic policy uncertainty indexes show similar fluctuations over time.

Another core explanatory variable is the digitalization level of commercial banks. In this paper, the digitalization level index of commercial banks is constructed based on the

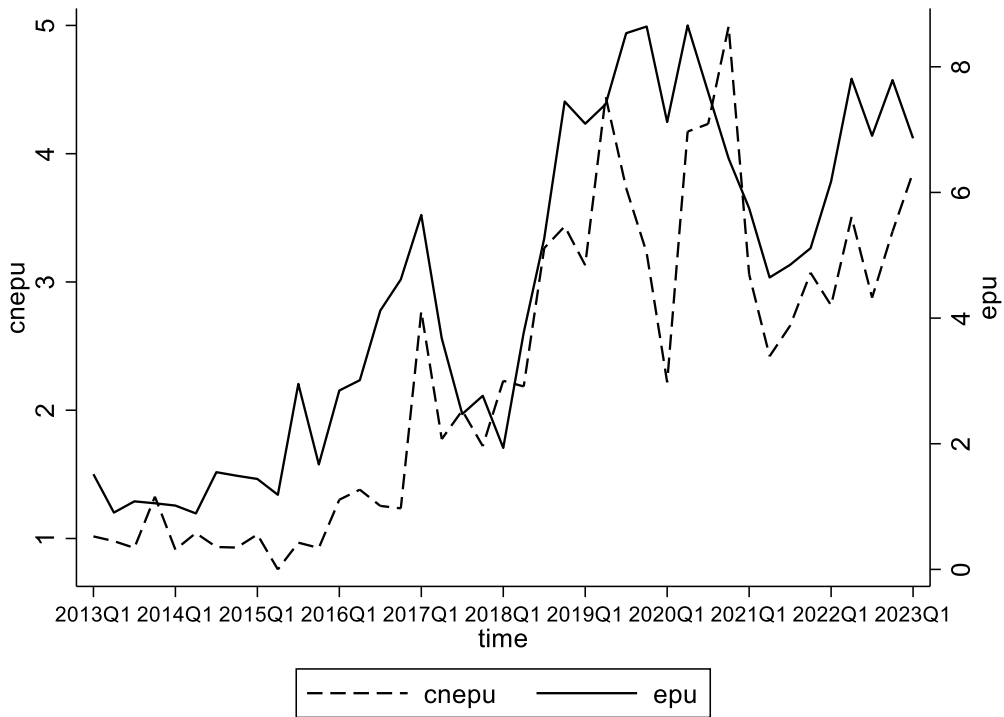


Figure 2. China economic policy uncertainty index, quarterly.

This figure depicts the China economic policy uncertainty index from the first quarter of 2013 to the first quarter of 2023. In this graph, *cnepu* represents the economic policy uncertainty index measured by Davis, Liu, and Sheng (2019) based on mainland China newspapers, *epu* represents the economic policy uncertainty index measured by Baker, Bloom, and Davis (2016) using text analysis. The data source is http://www.policyuncertainty.com/china_monthly.html.

number of Baidu News articles related to digitalization keywords. The rationale for using the quantity of Baidu News articles is that the digital practices, transformations, and developments of commercial banks are likely to attract media attention and thus be retrievable through Baidu News searches. In the era where the internet serves as the primary medium for information dissemination, the quantity of news articles is closely related to public attention (Askatas and Zimmermann 2009). The more attention a subject receives from the public, the more frequent the digital practices of commercial banks become, indicating a higher level of digitalization.

This paper first extracted keywords related to the digitalization of commercial banks from various sources, including the 'Guiding Opinions on the Digital Transformation of the Banking and Insurance Industry' issued by the China Banking and Insurance Regulatory Commission, the 'Development Plan for Fintech (2022–2025)' released by the People's Bank of China, and other relevant important news and conferences. These keywords were categorized into five dimensions: strategic planning, organizational structure, business processes, operational services, and data governance, see Appendix Table A1 for details.

Next, these keywords will be matched with commercial banks, such as 'China Agricultural Bank + Digitalization Strategy'. By using advanced search on Baidu News,

news articles that contain both 'China Agricultural Bank' and 'Digitalization Strategy' can be retrieved. Then, the web page source code of the advanced search results will be crawled, and the total number of news articles will be extracted. The natural logarithm of the sum of news article counts for all keywords related to the same bank will be used as the final indicator. Specifically, this paper limits the search time interval of news articles to the year 2016 to control for the initial digitalization heterogeneity of commercial banks. Considering that bank size may be positively correlated with the number of Baidu News articles, we introduce bank size as a control variable in the baseline regression. Additionally, we addressed the sample selection problem caused by bank size in the section of endogeneity treatment.

In the robustness test section, we use the digitalization level of listed banks (*digitallist*) and the availability of mobile banking applications (*app*) in commercial banks as alternative indicators. Firstly, the technological foundation of banking digitalization includes artificial intelligence, blockchain, cloud computing, and big data (Gomber, Koch, and Siering 2017). Based on these four dimensions, we construct a lexicon of feature words to represent the digitalization level (see Appendix Table A2). Web scraping techniques are employed to extract the frequency of words related to artificial intelligence, blockchain, cloud computing, and big data from annual reports to portray the digitalization level of listed commercial banks. Considering the differences in the length of annual report texts, the total word frequency for each listed commercial bank in the annual report is divided by the total number of words in the report. For convenience, the final indicator is multiplied by 10,000. Furthermore, mobile banking, as a digital infrastructure for banks, is also a reflection of their digitalization level (Core and De Marco 2021). Therefore, the availability of mobile banking services is used as an alternative indicator of the digitalization level. If a commercial bank offers mobile banking services, a binary variable is set to 1; otherwise, it is set to 0. The data on mobile banking availability are collected from bank websites and through phone consultations. Similarly, the data for these two alternative indicators are still based on the year 2016.

3.2.3. Control variables and subsample variables

Based on the studies by Berger et al. (2022), Nguyen (2021), Wu et al. (2020), and Wu et al. (2022), this paper introduces several control variables at the bank level, as follows: (1) Profitability (*ebitast*): Measured by the ratio of operating profit to total assets. (2) Operating efficiency (*opcopr*): Represented by the cost-to-income ratio, where a smaller value indicates higher efficiency. (3) Liquidity (*loandp*): Calculated as the ratio of total loans to total deposits, with a lower value indicating stronger liquidity. (4) Debt structure (*dplb*): Measured as the ratio of total deposits to total liabilities. (5) The natural logarithm of total assets (*lnast*), the natural logarithm of bank age (*lnage*), and the listed status (*list*) are used as control variables to respectively control for bank size, bank growth capacity, and overall strength.

Furthermore, this paper introduces macro-level control variables as follows: (1) Entity economic development level (*ggdp*): Represents the quarterly GDP growth rate. (2) Inflation level (*cpi*): Represented by the quarterly Consumer Price Index (CPI) for residents. (3) Changes in monetary policy (*intmarg*): Measured as the difference between long-term and short-term interest rates. (4) Banking industry concentration (*cr5*): Calculated as the

sum of total assets of the top five banks ranked by total assets, divided by the total assets of the banking industry.

In addition, this paper considers the following subsample variables for heterogeneity analysis: (1) Bank ownership (*ownership*): A binary variable with a value of 1 for large state-owned banks and 0 for other bank categories. (2) Pre-and-post pandemic period (*period*): A dummy variable with a value of 1 for quarters from the fourth quarter of 2019 to the third quarter of 2021, representing the period affected by the pandemic, and 0 for other quarters. (3) External environmental pressure (*fintech*): Measured by the provincial-level digital inclusive finance index from Peking University.⁵ Banks are divided into two groups based on the median value of the digital inclusive finance index in the province where they are located. Banks in provinces with high values are assigned a value of 1, and banks in provinces with low values are assigned a value of 0. It is important to note that all three variables are based on data from the year 2016.

3.3. Data sources and samples

The data used in this paper mainly includes four categories: commercial bank financial data, macroeconomic data, economic policy uncertainty index, and digitalization level data. In the first place, Quarter financial data of commercial banks were obtained from the Wind database, and missing values were supplemented by manually reviewing a large number of quarterly and annual reports of commercial banks. To avoid duplicate calculations, merged financial statements were consistently used during data download and collection. To mitigate the impact of outliers, samples with a lifespan of less than three quarters were excluded. Next, macroeconomic data were sourced from the Wind database and China National Bureau of Statistics. The third type of data comes from the official website of economic policy uncertainty index,⁶ and the fourth type of data is obtained through web scraping.

First, economic policy uncertainty exhibits significant short-term fluctuations, and due to sample capacity considerations, we use quarterly data. Second, we focus on whether digitalization level can contribute to bank stability under conditions of intensified economic policy uncertainty. Economic policy uncertainty reached its highest level since the first quarter of 2013 in the first quarter of 2017 (see Figure 2). Therefore, we select quarterly unbalanced panel data from 102 Chinese commercial banks from the first quarter of 2017 to the third quarter of 2021. Specifically, the sample includes 6 large state-owned banks, 11 joint-stock banks, 34 city commercial banks, and 51 rural commercial banks. As of the third quarter of 2021, the total assets of China's banking industry were 28.66⁷ trillion yuan, with the sample commercial banks' total assets reaching 23.13 trillion-yuan, accounting for 80.70% of the total banking industry assets. The sample represents a high level of representativeness. Table 1 displays the definitions and descriptive statistics of the main variables.

Before conducting regression analysis, we examined the relationship between the digitalization level of banks in the pre-sample period and their stability during the sample period. As a preview, Figure 3 shows the changes in bank stability for the bottom half and top half of the distribution of digitalization levels from the first quarter of 2017 to the third quarter of 2021. The data indicates that, in terms of overall trend, these two groups of banks are almost

Table 1. Descriptive statistics.

Variables	Definitions	N	St.Dev	Mean	Min	Max
Dependent variables						
<i>lnz</i>	Stability (Z-score)	1726	0.69	1.52	−2.65	2.80
<i>npl</i>	Non-performing loan ratio	1211	1.07	1.79	0.58	14.96
<i>aegoi</i>	Management expenses/Operating income	1726	0.22	0.35	0.12	6.90
<i>roa</i>	Return on Assets	1726	0.12	0.19	−0.24	0.74
<i>nii</i>	Non-interest income ratio	1703	20.11	27.86	−41.63	129.02
Core independent variables						
<i>cnepu</i>	Economic policy uncertainty index (Davis, Liu, and Sheng 2019)	1726	0.93	3.09	1.72	4.99
<i>epu</i>	Economic policy uncertainty index (Baker, Bloom, and Davis 2016)	1726	2.06	5.91	1.93	8.66
<i>digital</i>	Digitalization Level (Baidu News)	1726	0.38	7.74	6.79	8.48
<i>digitallist</i>	Digitalization level of listed commercial banks	418	0.62	1.21	0.45	2.90
<i>app</i>	Release mobile banking applications or not	1726	0.46	0.30	0	1
Bank-level control variables						
<i>ebitast</i>	Ratio of operating profit to total assets	1726	0.16	0.24	−1.10	0.91
<i>opcopr</i>	Operating costs/operating income	1726	0.09	0.31	0.12	0.77
<i>loandp</i>	Total loans/total deposits	1726	0.20	0.73	0.39	6.87
<i>dplb</i>	Total deposits/total liabilities	1726	0.12	0.76	0.38	1.06
<i>lnast</i>	Natural logarithm of total assets	1726	2.08	26.15	22.81	31.20
<i>lnage</i>	Natural logarithm of bank age	1726	0.57	2.71	1.39	3.64
<i>list</i>	Listed or unlisted	1726	0.48	0.35	0	1
Macro-level control variables						
<i>ggdp</i>	GDP growth rate	1726	4.62	9.14	−3.25	20.73
<i>cpi</i>	Consumer Price Index (CPI)	1726	1.23	2.04	−0.03	4.97
<i>intmarg</i>	Difference between long-term and short-term interest rates	1726	0.22	0.60	0.18	1.15
<i>cr5</i>	Sum of total assets of the top five banks ranked by total assets	1726	0.01	0.48	0.47	0.51
Subsample variables						
<i>ownership</i>	State-owned bank or not	1726	0.24	0.06	0	1
<i>period</i>	Post-epidemic period or not	1726	0.50	0.46	0	1
<i>fintech</i>	High fintech inclusive finance index province or not	1726	0.45	0.71	0	1

This table presents descriptive statistics and specific definitions of variables.

indistinguishable. However, considering the values of *lnz*, banks with high digitalization levels consistently exhibit higher stability than banks with low digitalization levels.

4. Empirical analysis results

4.1. Baseline regression

Table 2 presents the baseline regression results. Columns (1) to (3) report the regression results without introducing the interaction term of digitalization level. It can be observed that economic policy uncertainty has a significant negative effect on the stability of commercial banks, which is consistent with previous literature (Nguyen 2021; Wu et al. 2020, 2022). In columns (4) and (5), we introduce the bank's digitalization level, represented by the variable *digital*, and the interaction term $cnepu \times digital$. It can be observed that in these two columns, the regression coefficients for *digital*, and $cnepu \times digital$ are both significantly positive. Column (4) does not include bank-level control variables, while column (5) includes a subset of bank-level control variables. Since the level of digitalization of commercial banks varies only across individuals and not over time, we did not control for fixed effects in the regressions.

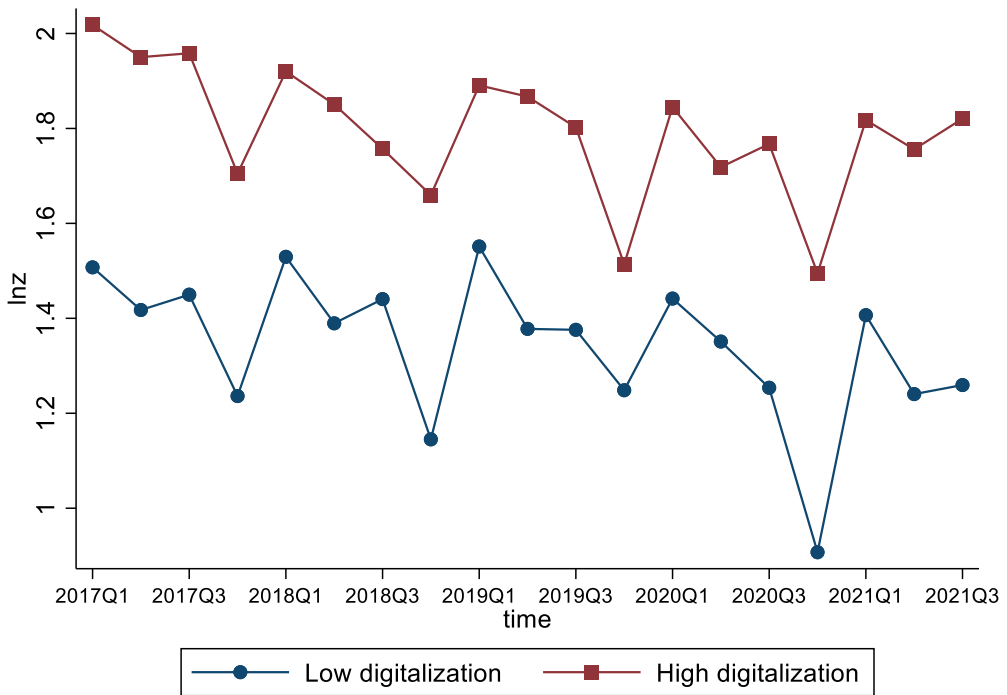


Figure 3. Stability of banks at different levels of digitization.

This figure illustrates the median stability of banks with high and low digitalization levels. 'High digitalization' refers to banks with a digitalization level above the 50th percentile of all banks, while 'low digitalization' refers to banks with a digitalization level below the 50th percentile of all banks. The specific time range includes from the first quarter of 2017 to the third quarter of 2021.

However, to eliminate or reduce the impact of individual characteristics that do not vary over time on the outcome variable, we control for bank-level individual fixed effects in the regressions for the other regressions. Columns (6) to (8) introduce the interaction term $cnepu \times digital$, along with the inclusion of bank-level control variables, macroeconomic control variables, and time fixed effects, respectively. The results show that the coefficient of the interaction term between economic policy uncertainty and digitalization level is consistently positive and statistically significant, validating hypothesis 1. Furthermore, even after controlling for time fixed effects, the coefficient of the interaction term $cnepu \times digital$ remains significantly positive, indicating that the digitalization level of commercial banks can indeed mitigate the negative impact of economic policy uncertainty on bank stability.

Based on the results in column (7) of the benchmark regression, when the level of digitalization of commercial banks is at its mean value, a one-unit increase in economic policy uncertainty will lead to a 3.2% decrease in the stability of commercial banks. However, if the level of digitalization of commercial banks increases by 10%, the stability of commercial banks will increase by 0.35%.⁸ Therefore, the digitalization level of commercial banks not only has statistical significance but also economic significance in significantly alleviating the adverse effects of macroeconomic policy uncertainty on bank stability, thereby ensuring stable operations of the banks.

Table 2. Baseline regression.

Dependent variable	<i>lnz</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>cnepu</i>	−0.064*** (−6.94)	−0.032*** (−4.21)	−0.032*** (−3.33)	−0.075*** (−6.42)	−0.045*** (−4.39)	−0.064*** (−7.07)	−0.032*** (−3.33)	
<i>digital</i>				0.573*** (5.39)	0.260** (2.08)			
<i>cnepu</i> × <i>digital</i>				0.056* (1.97)	0.051** (2.27)	0.053** (2.32)	0.035** (2.09)	0.035** (2.04)
<i>ebitast</i>		1.437*** (4.99)	1.430*** (4.98)		1.131*** (4.76)		1.425*** (4.96)	1.288*** (4.43)
<i>opcopr</i>		−0.850*** (−3.68)	−0.796*** (−3.51)		−1.862*** (−4.19)		−0.797*** (−3.51)	−0.479** (−2.30)
<i>loandp</i>		0.018 (1.09)	0.020 (1.29)		0.033 (0.23)		0.021 (1.39)	0.031* (1.81)
<i>lnage</i>		0.350*** (3.02)	0.334** (2.13)		0.375*** (4.19)		0.362** (2.29)	0.256 (1.56)
<i>dplb</i>		0.135 (0.68)	0.136 (0.73)				0.137 (0.73)	0.216 (1.25)
<i>lnast</i>		−0.152** (−2.16)	−0.127 (−1.60)				−0.134* (−1.69)	−0.012 (−0.16)
<i>list</i>		0.022 (0.38)	0.024 (0.40)				0.029 (0.49)	0.025 (0.42)
<i>ggdp</i>			0.003* (1.94)				0.003* (1.94)	
<i>cpi</i>			0.009 (1.09)				0.009 (1.11)	
<i>cr5</i>			0.751 (0.79)				0.892 (0.94)	
<i>intmarg</i>			0.086** (2.22)				0.083** (2.14)	
N	1,726	1,726	1,726	1,726	1,726	1,726	1,726	1,726
Adj. R ²	0.750	0.862	0.863	0.108	0.400	0.751	0.863	0.868
Bank FE	Y	Y	Y	N	N	Y	Y	Y
Quarter FE	N	N	N	N	N	N	N	Y

This table reports the results of baseline regression. The values in parentheses are robust t-statistics. The asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Specific definitions of variables are provided in Table 1.

4.2. Robustness checks and endogeneity

4.2.1. Indicator substitution

According to Wu et al. (2020), the impact of uncertainty gradually transforms into banking risk, leading to a gradual reduction in the stability of commercial banks. Considering the lagged effect of economic policy uncertainty, we introduce a one-period lag of the core explanatory variable *cnepu* and defines it as *lcnepu*. The regression results are presented in columns (1) and (2) of Table 3. Additionally, we utilize the economic policy uncertainty index *epu* proposed by Baker, Bloom, and Davis (2016) using text analysis. Similarly, the index is lagged by one period and defined as *lepu*, and the regression results are shown in columns (3) and (4) of Table 3.⁹ The coefficients of the lagged economic policy uncertainty and its interaction with the digitalization level show consistent signs and significance levels with the benchmark regression results, indicating the robustness of the research findings. It is worth noting that the significance level of the regression coefficient for *lepu* is relatively weaker. One possible explanation is that Baker, Bloom, and Davis (2016) provided the earliest precursor of the Chinese economic policy uncertainty index, primarily based on the

Table 3. Alternative indicators of economic policy uncertainty.

Dependent variable	<i>lnz</i>			
	(1)	(2)	(3)	(4)
<i>lcnepu</i>	-0.013** (-2.06)			
<i>lcnepu</i> × <i>digital</i>	0.036*** (3.46)	0.036*** (3.35)		
<i>lepu</i>			-0.002 (-0.30)	
<i>lepu</i> × <i>digital</i>			0.010** (2.08)	0.009* (1.84)
N	1,726	1,726	1,726	1,726
Adj. R ²	0.862	0.868	0.861	0.868
<i>X</i>	Y	Y	Y	Y
<i>M</i>	Y	N	Y	N
Bank FE	Y	Y	Y	Y
Quarter FE	N	Y	N	Y

This table presents the results of different measures of economic policy uncertainty. The values in parentheses are robust t-statistics. The asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Specific definitions of variables are provided in Table 1.

English-language newspaper ‘South China Morning Post’ from Hong Kong. In contrast, Davis, Liu, and Sheng (2019) built upon Baker, Bloom, and Davis (2016) work, incorporating mainland Chinese newspapers such as the ‘People’s Daily’ and ‘Guangming Daily’ and utilizing natural language processing techniques to select policy-related terms. This improvement represents a further refinement in measuring Chinese economic policy uncertainty.

Continuing, this paper employs alternative variables for digitization level, namely *digitallist* representing the level of digitization in listed commercial banks, and the dummy variable *app* indicating the presence of mobile banking services in commercial banks. Subsequently, these alternative variables are substituted for the core explanatory variable *digital* in the regression using Equation (3). The results are presented in Table 4, columns (1) and (2), as well as columns (3) and (4). Notably, even with the replacement of the digitalization measurement, the fundamental research conclusions of this paper remain unchanged.

4.2.2. Endogeneity treatment

As first, following Wooldridge (2010), lagging the explanatory variable by one period can help alleviate endogeneity issues arising from reverse causality. Therefore, this paper initially performs a regression with the explanatory variable lagged by one period, and the results are shown in columns (1) and (2) of Table 5. It can be observed that the coefficient’s sign and significance remain consistent with the benchmark regression. The conclusion remains robust.

Furthermore, to address the potential issue of stable banks being more likely to choose higher digitalization levels due to having more funds and stronger capabilities to enhance their digitalization, this paper adopts the approach proposed by Imbens and Wooldridge (2009) and updates the sample through a two-stage regression. In the first stage, we regress the commercial bank’s digitalization level as the dependent variable, controlling for bank stability (*lnz*), profitability (*ebitast*), total assets (*lnast*), growth capacity (*lnage*), listing status (*list*), and time fixed effects, while retaining the

Table 4. Alternative indicators of digitalization level.

Dependent variable	<i>lnz</i>			
	(1)	(2)	(3)	(4)
<i>cnepu</i>	−0.025 (−1.69)		−0.048*** (−3.11)	
<i>cnepu</i> × <i>digitallist</i>	0.021* (1.84)	0.019* (1.77)		
<i>cnepu</i> × <i>app</i>			0.028* (1.87)	0.027* (1.81)
N	418	418	1,726	1,726
Adj. R ²	0.954	0.958	0.863	0.868
<i>X</i>	Y	Y	Y	Y
<i>M</i>	Y	N	Y	N
Bank FE	Y	Y	Y	Y
Quarter FE	N	Y	N	Y

This table presents the results of different measures of digitalization level. The values in parentheses are robust t-statistics. The asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Specific definitions of variables are provided in Table 1.

Table 5. Explanatory variables lagged one-period regression and two-stage regression.

Dependent variable	<i>lnz</i>			
	(1)	(2)	(3)	(4)
<i>cnepu</i>	−0.087*** (−6.15)		−0.024*** (−3.19)	
<i>cnepu</i> × <i>digital</i>	0.054** (2.30)	0.069*** (2.98)	0.043** (2.13)	0.041** (2.05)
N	1,624	1,624	1,553	1,553
Adj. R ²	0.769	0.807	0.894	0.897
<i>X</i>	Y	Y	Y	Y
<i>M</i>	Y	N	Y	N
Bank FE	Y	Y	Y	Y
Quarter FE	N	Y	N	Y

This table displays the results of explanatory variables lagged one-period regression and two-stage regression. The values in parentheses are robust t-statistics. The asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Specific definitions of variables are provided in Table 1.

fitted values from the regression. Subsequently, we divide the 102 commercial banks into high and low digitalization level groups based on the median value of digitalization level. To address the endogeneity issue more effectively, we remove the top 10% of fitted values for high digitalization level group and the bottom 10% of fitted values for low digitalization level group, aiming to eliminate the most severe cases of reverse causality in the sample. Finally, in the second stage regression, we conduct the regression on the retained sample according to Equation (3), and the results are shown in columns (3) and (4) of Table 5. It indicates that even after considering the reverse causality issue, the sign and significance of the interaction term between economic policy uncertainty and digitalization level remain consistent. The persistent mitigating effect of digitalization level is evident after updating the optimized sample, confirming the robustness of the findings.

Thirdly, this paper also employs instrumental variable regression, following the approach of Liang, Zhao, and Huang (2023). For this purpose, the number of fixed telephones per hundred people at the city level in 1984 is preliminarily selected as an instrumental variable for the commercial bank's digitalization level. On one hand, the communication level in the city where a bank is located, based on the historical

development, can influence the bank's digitalization level from various aspects such as technological advancements and social preferences, satisfying the relevance condition. On the other hand, fixed telephones, as a social infrastructure, primarily provide daily communication services to the public and do not directly affect the stability of commercial banks, satisfying the exogeneity condition. Considering that this paper mainly focuses on the coefficient of the interaction term $cnepu \times digital$, the interaction between the economic policy uncertainty index and the number of fixed telephones per hundred people at the city level in 1984 is used as an instrumental variable for $cnepu \times digital$, and it is defined as iv . Column (1) of Table 6 reports the results of the first-stage regression, and the second-stage regression results are reported in Column (2). The Kleibergen-Paap rk LM statistic rejects the null hypothesis of instrument under-identification at the 1% significance level, and the Cragg-Donald Wald F statistic is much larger than the critical value of 24.58 for the weak instrument identification test at the 10% significance level, strongly rejecting the null hypothesis of weak instrument. Hence, the instrumental variables chosen in this paper are considered reasonable. These findings reaffirm the main research conclusions.

Furthermore, following the approach of S. Chen and Zhang (2021), we also use internet data as an instrumental variable. Specifically, the internet penetration rate at the provincial level in 2016 was employed, calculated as the number of internet users divided by the total population. This internet penetration rate was combined with the interaction term of economic policy uncertainty to create the instrumental variable $cnepu \times digital$, and it is also defined as iv . Similar to the previous instrumental variable, the relevant statistical tests rejected the null hypothesis of weak instrument and instrument insufficiency. Columns (3) and (4) of Table 6 present the first and second-stage regression results using the new instrumental variable. The coefficient of $cnepu \times digital$ remains statistically significant and positive.¹⁰ These findings reaffirm the reliability of the main conclusions.¹¹

Table 6. Instrumental variables regression.

Instrumental variables	Fixed telephone		Internet penetration	
	First stage $cnepu \times digital$ (1)	Second stage lnz (2)	First stage $cnepu_digital$ (3)	Second stage lnz (4)
Dependent variables				
$cnepu$	-0.001 (-0.03)	-0.035*** (-3.49)	-0.006 (-0.19)	-0.032*** (-3.31)
iv	0.204*** (7.71)		0.019*** (7.05)	
$cnepu \times digital$		0.057** (2.49)		0.051* (1.72)
N	1,608	1,608	1,726	1,726
Adj. R ²	0.327	0.483	0.286	0.466
X	Y	Y	Y	Y
M	Y	Y	Y	Y
Bank FE	Y	Y	Y	Y
Quarter FE	N	N	N	N
Kleibergen-Paap rk LM statistic		59.44***		49.68***
Cragg-Donald Wald F statistic		817.95		750.83

This table reports the results of instrumental variable regression. The values in parentheses are robust t-statistics. The asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Specific definitions of variables are provided in Table 1.

4.3. Mechanism analysis

After implementing the indicators replacement and addressing endogeneity concerns, we have obtained a robust and reliable core finding: digitalization can mitigate the negative impact of economic policy uncertainty on the stability of commercial banks. However, the specific mechanisms underlying this effect remain unclear. In this section, we attempt to address this question. Based on the theoretical analysis and mechanism discussion in the previous sections, we separately regress the non-performing loan ratio (*npl*), the ratio of administrative expenses to operating income (*aegoi*), and the return on assets (*roa*) of commercial banks as dependent variables against the right-hand side of Equation (3) as the explanatory variable. The regression results are reported in Table 7. In columns (1) and (2), an increase in economic policy uncertainty leads to a rise in the non-performing loan ratio of commercial banks, and digitalization significantly mitigates the negative impact of economic policy uncertainty on this ratio. In columns (3) and (4), economic policy uncertainty increases the management costs of risk management for commercial banks, and digitalization effectively alleviates the rise in risk management costs caused by economic policy uncertainty. In columns (5) and (6),¹² economic policy uncertainty significantly suppresses the profitability of commercial banks, and digitalization efficiently alleviates the decline in profitability caused by economic policy uncertainty.

In summary, firstly, when faced with heightened economic policy uncertainty, the digitalization level of commercial banks enhances their borrower screening capability through digital credit assessment, thereby reducing the non-performing loan ratio and improving stability, confirming hypothesis **H1a**. Secondly, as economic policy uncertainty rises, the digitalization level of commercial banks improves their ability for digital information analysis, enhancing their risk identification capability and consequently lowering risk management costs, thus improving bank stability, validating hypothesis **H1b**. Thirdly, with the increase in economic policy uncertainty, digital operational services under digitalization provide profit security for commercial banks, thereby increasing the return on total assets and enhancing bank stability, confirming hypothesis **H1c**. Therefore, digitalization indeed helps reduce the risk and bankruptcy probability of commercial banks, ensuring their stability and resilience against the adverse impact of economic policy uncertainty.

Table 7. Mechanism analysis regression.

Dependent variables	<i>npl</i>		<i>aegoi</i>		<i>roa</i>	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>cnepu</i>	0.004 (0.58)		0.012 (1.25)		−0.015*** (−5.27)	
<i>cnepu</i> × <i>digital</i>	−0.064*** (−3.13)	−0.074*** (−3.49)	−0.025* (−1.79)	−0.025* (−1.73)	0.016** (2.52)	0.015** (2.26)
N	1,209	1,209	1,726	1,726	1,726	1,726
Adj. R ²	0.992	0.993	0.268	0.276	0.319	0.399
<i>X</i>	Y	Y	Y	Y	Y	Y
<i>M</i>	Y	N	Y	N	Y	N
Bank FE	Y	Y	Y	Y	Y	Y
Quarter FE	N	Y	N	Y	N	Y

This table reports the results of mechanism analysis regression. The values in parentheses are robust t-statistics. The asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Specific definitions of variables are provided in Table 1.

4.4. Heterogeneity analysis

After clarifying the role of digitalization, we further examines whether this influence varies across different banks, time periods, and external environments. Regarding the heterogeneity analysis, we select three variables: ownership (whether the bank is state-owned or not), period (whether it is the post-pandemic period), and fintech (whether the province is classified as a high digital inclusive finance index province) for sub-sample regressions. The regression results are presented in Table 8.¹³

First, Berger et al. (2005), based on incomplete contract theory and data from U.S. banks, found that large banks are less willing to lend to firms without financial records. Berger and Black (2011) also pointed out that large banks tend to lend to relatively larger and more transparent firms, while small banks primarily serve smaller and less transparent firms. Therefore, large banks mainly target high-quality customers with good financial information, while small banks mainly serve lower-quality customers. As a result, in the case of increased economic policy uncertainty, the likelihood of an increase in non-performing loans is higher for small banks. Consequently, they would be more in need of the mitigating effect of digitalization compared to large banks. State-owned banks, also known as large state-owned banks, usually have larger scale and stronger overall capabilities. Thus, this paper divides the sample into two groups based on whether the banks are state-owned or not and performs sub-sample regressions. Columns (1) and (2) present the regressions for the sample of non-state banks and the sample of state-owned banks, respectively. It is evident that the coefficient of the interaction term $cnepu \times digital$ is more significant in the non-state-owned bank sample, which aligns with our expectations and further validates the research hypothesis H1a.

Moreover, the outbreak of the COVID-19 pandemic at the end of 2019 further intensified economic policy uncertainty (Baker et al. 2020). Consequently, banks faced even greater information asymmetry and experienced a more significant decline in risk identification ability compared to the pre-pandemic period. This led to higher risk management costs, making them more in need of the mitigating effect of digitalization to reduce these costs. Therefore, this paper divides the sample into pre-pandemic and post-pandemic periods to investigate the impact of the pandemic on the relationship between economic

Table 8. Heterogeneity analysis regression.

	Non-state-owned	State-owned	Pre <i>lnz</i>	Post	Low	High
Dependent variables	(1)	(2)	(3)	(4)	(5)	(6)
<i>cnepu</i>	−0.032*** (−3.16)	−0.011 (−1.00)	−0.008 (−0.26)	−0.085*** (−3.09)	−0.063*** (−4.07)	−0.024* (−1.91)
<i>cnepu</i> × <i>digital</i>	0.038** (1.99)	0.025 (1.13)	0.035* (1.90)	0.058* (1.79)	−0.012 (−0.32)	0.037* (1.73)
N	1,618	108	935	788	497	1,229
Adj. R ²	0.856	0.961	0.847	0.886	0.835	0.862
<i>X</i>	Y	Y	Y	Y	Y	Y
<i>M</i>	Y	Y	Y	Y	Y	Y
Bank FE	Y	Y	Y	Y	Y	Y
Quarter FE	N	N	N	N	N	N

This table reports the results of heterogeneity analysis regression. The values in parentheses are robust t-statistics. The asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Specific definitions of variables are provided in Table 1.

policy uncertainty and digitalization. Columns (3) and (4) are pre-epidemic sample regressions, post-epidemic sample regressions, respectively. It is observed that the coefficient of the interaction term $cnepu \times digital$ is more significant in the post-pandemic sample, which aligns with our expectations and further strengthens the research hypothesis **H1b**.

Finally, the existence of competitive effects can also lead to varying impacts of digitalization on bank stability under uncertainty. With the rapid development of fintech, it has significantly affected traditional financial and banking industries (Gomber, Koch, and Siering 2017). In provinces where third-party fintech is well-developed, commercial banks located in those regions face greater competitive pressure, and their profitability can be influenced by third-party fintech companies. In the context of intensified economic policy uncertainty, these banks would be in greater need of the mitigating effect of digitalization. Considering the heterogeneous mitigating effect of digitalization under different external competitive environments, this paper uses the provincial-level fintech inclusion index measured by the Peking University Digital Finance Research Center to classify commercial banks into high and low fintech inclusion index groups based on their provinces. Subsequently, the sample is divided, and regressions are conducted for each group. The fifth and sixth columns represent the regressions for the low and high fintech inclusion index groups, respectively. It is observed that the coefficient of the interaction term $cnepu \times digital$ is more significant in the high fintech inclusion index group, which aligns with our expectations and further strengthens the research hypothesis **H1c**.

5. Conclusion

This paper utilizes quarterly data from Chinese commercial banks to explore the stability performance of banks with different levels of digitalization under economic policy uncertainty. The findings indicate that banks with higher digitalization levels exhibit better stability when facing intensified economic policy uncertainty. Digitalization achieves this by utilizing transmission channels between economic policy uncertainty and bank stability, such as leveraging digital credit assessment for borrower selection, benefiting from lower risk management costs through digital information analysis, and ensuring profitability through digital service operations. Therefore, digitalization brings gains to bank stability in the context of economic policy uncertainty and ensures the stable operation of banks.

This paper examines the digitalization and stable development of the banking industry in China as a representative developing country, offering crucial insights for digital transformation and financial stability in developing economies. Firstly, governments in developing countries should scientifically formulate and implement digital transformation strategies to reduce construction costs. While digitalization is inevitable, governments need to adopt differentiated strategies tailored to different banks, external environments, and periods, especially focusing on enhancing the digitalization level of small banks and those facing high external competitive pressures. In the post-pandemic era, governments should not only strive to enhance the digitalization level of commercial banks but also focus on the overall stability of the banking industry, preventing financial risks, balancing financial innovation with financial security, and alleviating the impact of economic policy uncertainty on bank stability. Secondly, banks in developing countries should pay particular attention to digital credit assessment capabilities, digital information analysis capabilities, and digital

operational service capabilities during the digitalization process, as these are vital channels for leveraging the mitigating effects of digitalization. Through steady digital transformation of the banking sector, developing countries' commercial banks can better cope with the challenges posed by economic policy uncertainty, enhance the stability of the financial system, and promote economic growth and sustainable development.

Finally, future research could consider cross-country comparisons to explore how digitalization levels influence the relationship between economic policy uncertainty and banking stability across different countries or regions. For instance, examining cross-country samples from multiple developing countries could reveal differences in bank responses to economic policy uncertainty shocks. This would further investigate the experiences and lessons learned from the digital transformation and financial development of the banking sector in various countries, providing valuable insights and guidance for the digital transformation and financial growth of banks in developing countries. This is a promising area for future research that can offer broader policy recommendations.

Notes

1. According to the annual performance reports disclosed by various banks in China, in 2022, the total investment in fintech by six large state-owned commercial banks and ten national joint-stock commercial banks amounted to 178.764 billion yuan, an increase of 14.204 billion yuan compared to the previous year, representing a year-on-year growth of 8.63%. In terms of the proportion of fintech to operating income, it mainly ranged from 2.83% to 5.27%. Among them, thirteen banks saw an increase in the proportion of fintech to operating income compared to 2021.
2. China's largest search engine, as a leading Chinese search engine, holds an almost monopolistic dominant position in the search engine market in China. This further illustrates the comprehensiveness of news quantity coverage and the reliability of measuring digitalization level.
3. The change of $100\beta_1^*\%$ is since the stability of commercial banks is measured in natural logarithms and is ultimately defined as $\ln z$.
4. The calculation of $\sigma(ROA)$ (standard deviation of Return on Assets) in this article is based on the entire sample period. One reason for this approach is to include as much information as possible. Additionally, using the full sample period helps to prevent any loss of data.
5. Data source: Beijing University Digital Finance Research Center (<https://www.idf.pku.edu.cn/>).
6. Data source: http://www.policyuncertainty.com/china_monthly.html.
7. The data is from the official website of the People's Bank of China (<http://www.pbc.gov.cn/en/>).
8. The use of percentage change to interpret the regression coefficients is since both the stability of commercial banks and the level of digitalization are measured in natural logarithms in this paper.
9. In addition, we also utilized the China Economic Policy Uncertainty Index measured by Y. Huang and Luk (2020). The main qualitative research conclusions remain consistent and hold true.
10. The economic significance of the coefficients in both instrumental variable regressions has increased, possibly due to the 'Local Average Treatment Effect' (LATE). Instrumental variable regression may capture the average treatment effect for only some specific individuals or subgroups in the sample, rather than the average treatment effect for all individuals.
11. The regression results using the two-way fixed effects model are also robust and consistent with the main findings. However, due to space constraints, they were not reported.
12. Because the dependent variable in the regressions for the third and fourth columns is the profitability of commercial banks (*roa*), the variable *ebitast*, which is another measure of

commercial bank profitability, is not included as a control variable at the bank level in this paper.

13. The regression results of the two-way fixed effects model, controlling for macro-level control variables, are consistent with the results of the individual fixed effects model. However, due to space constraints, they are not reported here.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

My paper is supported by the “2022 New Era Talent Cultivation Provincial Quality Engineering Project - Provincial Graduate Student Academic Innovation Project (2022xscx013).”

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Appendices

Table A1. Commercial bank digitization level keywords.

Dimensions	keywords
Strategic planning	Digitalization strategy, fintech strategy, digital banking, intelligent banking.
Organizational structure	Fintech department, fintech subsidiary, electronic banking department, digital professionals.
Business processes	Digital marketing, Robo-Advisory, one-stop service, integrated scene service platform.
Operational services	Mobile banking, online banking, smart branch, WeChat banking.
Data governance	Big data governance mechanism, intelligent data governance platform, data standardization system, data middle platform architecture.

Table A2. Feature words s of digitization level of listed commercial banks.

Dimensions	Feature words
Artificial intelligence	Artificial intelligence, image understanding, intelligent data analysis, intelligent robots, machine learning, deep learning, semantic search, biometric technology, facial recognition, speech recognition, identity verification, natural language processing, real-time monitoring, fingerprint recognition.
Blockchain	Blockchain, digital currency, distributed computing, differential privacy technology, intelligent finance, public chain, private chain, consortium chain, side chain, interchain.
Cloud computing	Cloud computing, cloud architecture, cloud services, cloud finance, stream computing, graph computing, in-memory computing, multi-party secure computing, brain-like computing, green computing, cognitive computing, converged architecture, billion-level concurrency, EB-level storage, internet of things (IoT), cyber-physical systems (CPS).
Big data	Big data, data stream, data layer, dataset, data mining, text mining, data visualization, heterogeneous data, credit reporting, augmented reality, mixed reality, virtual reality.